KG COLLEGE OF ARTS AND SCIENCE
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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

Course Code	Course Name	Category	Hours/Week	Credits
25IDT2AE	Innovation and Design Thinking	AECC-II	2	2

Question Bank

UNIT -1

1. **The primary focus of Design Thinking is to shift attention from internal business requirements to:**
   a) Technological capabilities b) The human beings using the solution
   c) Economic viability d) Competitor analysis
   Answer: b) The human beings using the solution

2. **Usability is best defined as:**
   a) The aesthetic appeal of a product
   b) The technical code behind an interface
   c) How easily and effectively a user can achieve a desired outcome
   d) The price point of a product
   Answer: c) How easily and effectively a user can achieve a desired outcome

3. **Which of the following is NOT one of the five core components of usability?**
   a) Learnability b) Efficiency c) Creativity d)Satisfaction
   Answer: c) Creativity

4. **"Learnability" refers to:**
   a) How fast experienced users perform tasks
   b) How quickly new users understand the product
   c) How easily users recover from errors
   d) How memorable the design is after a break
   Answer: b) How quickly new users understand the product

5. **"Efficiency" in usability focuses on**
   a) First-time users
   b) The speed at which experienced users perform tasks
   c) The visual layout of the menu
   d) Minimizing the cost of production
   Answer: b) The speed at which experienced users perform tasks

6. **If a user returns to an app after six months and can instantly remember how to use it, the app has high**
   a) Memorability b) Satisfaction c) Feasibility d) Learnability
   Answer: a) Memorability

7. **Good "Error Management" in design means**
   a) Users never make mistakes
   b) The system hides errors from the user
   c) Errors are few, and recovery is easy and clear
   d) The system crashes to prevent wrong data entry
   Answer: c) Errors are few, and recovery is easy and clear

8. **Which attribute of usability bridges the gap between mechanics and the broader User Experience (UX)?**
   a)Efficiency b)Errors c)Satisfaction d)Memorability
   Answer: c) Satisfaction

9. **In the Airbnb case study, what was identified as the primary reason listings were "unusable"?**
   a) High prices b) Poor quality, amateur photos
   c) Confusing website navigation d) Lack of reviews
   Answer: b) Poor quality, amateur photos

10. **What "non-scalable" action did the Airbnb founders take to fix their problem?**
    a) They rewrote the entire code base
    b) They lowered all prices by 50%
    c) They personally went to New York to take professional photos
    d) They launched a massive TV ad campaign
    Answer: c) They personally went to New York to take professional photos

11. **Changing the Airbnb "Star" icon to a "Heart" icon increased engagement by 30% because**
    a) It was easier to click
    b) It matched the "aspirational" mental model of users
    c) Hearts are a universal symbol of medical health
    d) The star icon was broken
    Answer: b) It matched the "aspirational" mental model of users

12. **Human-Centeredness (HCD) dictates that innovation must begin with**
    a) A new technology
    b) A business goal
    c) Understanding the people who will use the solution
    d) A financial audit
    Answer: c) Understanding the people who will use the solution

13. **In the "Three Lenses of Innovation" HCD primarily focuses on**
    a) Feasibility b) Viability c) Desirability d) Profitability
    Answer: c) Desirability

14. "Feasibility" asks the question

a) Do people want this? b) Is this financially sustainable?
c) What is technically and organizationally possible? d) How will this impact society?

Answer: c) What is technically and organizationally possible?

15. Which role does a Curator play in the design process?

a) Developing user interfaces
b) Selecting the most relevant content or ideas
c) Testing prototypes with users
d) Creating the final product.

Answer: b) Selecting the most relevant content or ideas

16. In the Mobile Kunji case study, why did rural families mistrust the ASHAs?

a) They charged too much money
b) They were seen as "just another local woman" without authority
c) They didn't speak the local dialect
d) They were always late

Answer: b) They were seen as "just another local woman" without authority

17. What insight about "voice" was critical in the Mobile Kunji design?

a) People preferred reading text over listening
b) A recorded voice from a machine carried more weight/authority than a neighbour
c) The ASHAs needed to shout to be heard
d) Music was more effective than speech

Answer: b) A recorded voice from a machine carried more weight/authority than a neighbour

18. Who was "Dr. Anita" in the Mobile Kunji solution?

a) The lead designer of the project
b) A fictional, authoritative audio character who gave medical advice
c) The head of the village council
d) A real doctor who visited every village

Answer: b) A fictional, authoritative audio character who gave medical advice

19. Empathy is described as the _____ of the Design Thinking process.

a) Final step b) Financial goal
c) Cornerstone and starting point d) Marketing strategy

Answer: c) Cornerstone and starting point

20. The primary goal of the "Empathize" stage is to

a) Sell the product to the user
b) Set aside personal assumptions to see the world through the user's eyes
c) Prototype a solution quickly
d) Calculate the budget

Answer: b) Set aside personal assumptions to see the world through the user's eyes

21. Which of the following is a method used to build empathy?

a) A/B Testing b) Cost-benefit analysis
c) Observation/Shadowing d) Compiling sales statistics

Answer: c) Observation/Shadowing

22. In the Oral-B toothbrush case study, the initial incorrect assumption was

a) Children need brighter colours b) Children are just "small adults"
b) Toothbrushes should be electric d) Kids don't like brushing their teeth

Answer: b) Children are just "small adults"

23. Observation revealed that small children hold toothbrushes using

a)Their fingertips b)A "fist grip" c)Two hands d)Only their thumbs

Answer: b) A "fist grip"

24. The solution for the Oral-B children's toothbrush was to

a) Make the handle thinner b) Make the bristles harder
c) Create a "Squish Grip" with a fat, soft handle d) Add a timer to the brush

Answer: c) Create a "Squish Grip" with a fat, soft handle

25. What is the main objective of Journey Mapping?

a) Identify customer archetypes
b) Understand the customer's experience over
c) Develop a new product concept
d) Improve internal business operations

Answer: b) Understand the customer's experience over

26. In the "What? How? Why?" framework, the "Why" column represents

a) The concrete observation b) The description of the action
c) The emotional inference d) The time of day

Answer: c) The emotional inference

27. Why should designers interview "Extreme Users"?

a) They are the easiest to find
b) They highlight pain points that average users often ignore or work around
c) They are the most profitable customers
d) They confuse the design process less

Answer: b) They highlight pain points that average users often ignore or work around

28. The "5 Whys" technique is used to

a) Get to the root emotional need or cause of a problem
b) Annoy the user
c) Check 5 different features of a product
d). Interview 5 different people simultaneously

Answer: a) Get to the root emotional need or cause of a problem

29. Iteration is best described as

a) A linear, step-by-step process that never repeats
b) A cyclical process of refining and testing based on feedback
c) Planning everything perfectly before building
d) The final marketing launch of a product

Answer: b) A cyclical process of refining and testing based on feedback

30. The core mindset of iteration is

a) "Get it right the first time" b) "Fail fast, learn faster"
c) "Avoid failure at all costs" d) "Build it and they will come"

Answer: b) "Fail fast, learn faster"

31. The iterative loop primarily connects which two stages of Design Thinking?

a) Empathize and Define b) Prototype and Test
c) Ideate and Empathize d) Define and Prototype

Answer: b) Prototype and Test

32. In the door handle example, iteration fixed the confusion between

a) Locking and unlocking b) Pushing and Pulling
c) Entrance and Exit d) Wood and Metal materials

Answer: b) Pushing and Pulling

33. Product Innovation focuses on

a) Internal workflows and supply chains
b) The creation or enhancement of goods/services offered to the market
c) Changing the revenue model
d) Solving societal issues

Answer: b) The creation or enhancement of goods/services offered to the market

34. Process Innovation aims to

a) Improve internal efficiency and reduce costs b) Create a new physical gadget
c) Change how the customer pays d) Increase the price of the product

Answer: a) Improve internal efficiency and reduce costs

35. A company shifting from selling software as a one-time purchase to a monthly subscription (SaaS) is an example of

a) Product Innovation b) Process Innovation
c) Business Model Innovation d) Social Innovation

Answer: c) Business Model Innovation

36. Social Innovation is distinct because its primary measure of success is

a) Maximum private profit b) Social outcomes and public benefit
c) Technological novelty d) Market domination

Answer: b) Social outcomes and public benefit

37. In the Wello Water Wheel case study, the "water burden" in rural India primarily affected

a) Men and the elderly b) Women and children
c) Farmers and merchants d) Government officials

Answer: b) Women and children

38. The Wello team redefined the problem from "lack of water" to

a) Lack of pumps b) The physical and time-intensive burden of water transport
c) The cost of water d) The cleanliness of the wells

Answer: b) The physical and time-intensive burden of water transport

39. The Wello Water Wheel allowed users to carry water by

a) Rolling it b) Using a backpack
c) Using a motorized scooter d) Pumping it through a hose

Answer: a) Rolling it

40. How much water could the Waterwheel hold compared to a traditional head-load?

a) The same amount b) Less water
c) 2.5x more water d) 10x more water

Answer: c) 2.5x more water

41. What was a major "Social" impact of the Waterwheel regarding gender roles?

a) Men were willing to help because rolling was seen as "using a tool"
b) Women stopped fetching water entirely
c) Children were banned from fetching water
d) It created a new job exclusively for men

Answer: a) Men were willing to help because rolling was seen as "using a tool"

42. According to the text, a product with high usability feels

a) Expensive b) Complex c)Intuitive d) Heavy

Answer: c) Intuitive

43. Which lens of innovation corresponds to "Business Sustainability"?

a) Desirability b) Feasibility c) Viability d)Usability

Answer: c) Viability

44. "Jugaad with Empathy" in the Indian context implies

a) Using cheap materials regardless of quality
b) Designing for social hierarchy and delivery medium, not just information
c) Copying western designs exactly
d) Ignoring the user's needs to save money

Answer: b) Designing for social hierarchy and delivery medium, not just information

45. Which Design Thinking stage involves creating a "Minimum Viable Product" (MVP)?

a) Empathize b) Define c)Prototype d) Test

Answer: c) Prototype

46. In the Airbnb example, the "Snow White Project" involved creating

a) A movie about the founders
b) Storyboards for the entire guest journey
c) A new logo based on a fairy tale
d) A specialized camera for hosts

Answer: b) Storyboards for the entire guest journey

47. A "low-fidelity prototype" is characterized by being

a) Expensive and polished
b) Quick, cheap, and rough (e.g., sketches)
c) Ready for mass manufacturing
d) Coded perfectly

Answer: b) Quick, cheap, and rough

48. Why is "Time-Poverty" a crucial factor in Social Innovation?

a) Poor people have too much free time
b) Saving time can be a ladder out of poverty
c) Social projects must be completed in one day
d) Watches are expensive

Answer: b) Saving time can be a ladder out of poverty

49. The "Circular Economy" is an example of which type of innovation?

a) Product Innovation
b) Process Innovation
c) Business Model Innovation
d) Social Innovation

Answer: d) Social Innovation

50. The "Define" stage of Design Thinking typically follows which stage?

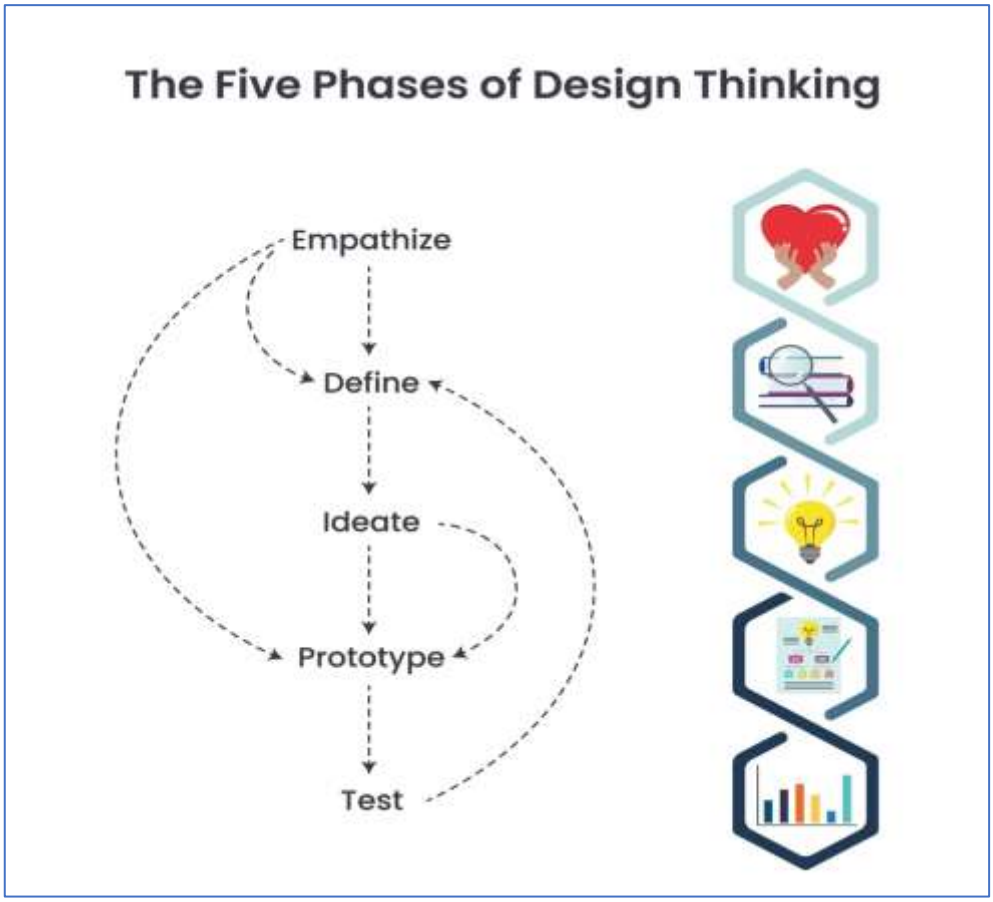
a) Ideate
b) Prototype
c) Empathize
d) Test

Answer: c) Empathize

UNIT – 2

1. **Empathy in Design Thinking is best described as**
   a) Feeling sorry for the user
   b) A critical, investigative practice of understanding the user's experience
   c) A method for analysing market competitors
   d) A process of writing code for the user

   Answer: b) A critical, investigative practice of understanding the user's experience



2. **Which of the following is NOT a primary function of Empathy in design?**
   a) Uncovering unarticulated needs b) Defining the right problem
   c) Maximizing immediate business profits d) Building solutions for real human problems

   Answer: c) Maximizing immediate business profits

3. **"Latent needs" are defined as**
   a) Needs that users loudly complain about
   b) Problems users have learned to tolerate and often cannot articulate
   c) Requirements explicitly written in a project brief
   d) Future technology trends

   Answer: b) Problems users have learned to tolerate and often cannot articulate

4. **A human-centered problem statement is also known as a**
   a) Business Plan b) Point of View (POV)
   c) Market Strategy d) Profit Margin Goal
   Answer: b) Point of View (POV)

5. **Which empathy tool involves asking open-ended questions to elicit emotional context?**
   a) Immersion b) Deep Interviews
   c) Shadowing d) A/B Testing
   Answer: b) Deep Interviews

6. **If a user says their experience is "fine" but their story reveals frustration, this insight was likely gained through**
   a) A survey b) A Deep Interview
   c) A focus group d) Quantitative data analysis
   Answer: b) A Deep Interview

7. **"Direct Observation" or "Shadowing" involves**
   a) Asking the user to write a diary
   b) Watching users in their natural environment without interruption
   c) Sending a questionnaire to 1,000 people
   d) Guessing what the user does
   Answer: b) Watching users in their natural environment without interruption

8. **Watching a healthcare worker struggle with small buttons while wearing gloves is an example of**
   a) Direct Observation b) Market Segmentation
   c) Ideation d) Prototyping
   Answer: a) Direct Observation

9. **"Immersion" in Design Thinking means**
   a) The design team physically experiences the user's situation
   b) Reading books about the user
   c) Watching videos of the user
   d) Hiring a consultant
   Answer: a) The design team physically experiences the user's situation

10. **Mind Mapping and Brainstorming sessions are designed to:**
    a) Visualize customer personas b) Generate and organize ideas for problem-solving
    c) Map the competitive landscape d) Analyse the value chain
    Answer: b) Immersion

11. **An Empathy Map synthesizes research by mapping which four elements?**
    a) Strengths, Weaknesses, Opportunities, Threats
    b) Says, Thinks, Does, Feels
    c) Who, What, Where, When
    d) Product, Price, Place, Promotion
    Answer: b) Says, Thinks, Does, Feels

12. In an Empathy Map, "Thinks" refers to

a) Direct quotes from the user
b) Unspoken thoughts or beliefs inferred from behaviour
c) The user's physical actions
d) The user's job title

Answer: b) Unspoken thoughts or beliefs inferred from behaviour

13. In the "Define" stage, empathy helps transform observations into:

a) A marketing slogan b) A human-centered problem statement (POV)
c) A financial spreadsheet d) A final product

Answer: b) A human-centered problem statement (POV)

14. During the "Ideate" stage, empathy serves as

a) A distraction
b) Guardrails to keep ideas focused on human needs
c) A method to reject all new ideas
d) A tool for cost reduction

Answer: b) Guardrails to keep ideas focused on human needs

15. During the "Prototype" stage, empathy reminds the team to focus on

a) High-fidelity graphics b)User interaction and experience
c) Perfect coding d) Expensive materials

Answer: b) User interaction and experience

16. When a user struggles during the "Test" stage, an empathetic team will

a) Blame the user
b) Become defensive about their design
c) Interpret the feedback objectively and iterate
d) Ignore the feedback

Answer: c) Interpret the feedback objectively and iterate

17. The core principle of understanding users goes beyond demographics to uncover

a) Emotional context and motivations b) Their annual income only
c) Their zip code d) Their favourite colour

Answer: a) Emotional context and motivations

18. Ethnographic research is drawn from the field of

a) Economics b) Anthropology c) Physics d) Mathematics

Answer: b) Anthropology

19. Ethnographic research requires the researcher to

a) Send out email surveys
b) Embed themselves in the user's culture or environment
c) Analyse stock market trends
d) Conduct phone interviews from the office

Answer: b) Embed themselves in the user's culture or environment

20. The primary focus of Ethnographic research is to understand

a) How much money the user has
b) The context and culture surrounding the user's problem
c) The technical specifications of a product
d) The competitor's market share

Answer: b) The context and culture surrounding the user's problem

21. In Design Thinking interviews, it is best to ask

a) Yes/No questions b) Open-ended questions that encourage storytelling
c) Leading questions d) Rhetorical questions

Answer: b) Open-ended questions that encourage storytelling

22. The "5 Whys" technique is used during interviews to

a) Annoy the user b) Uncover root motivations
c) Finish the interview quickly d) Confuse the user

Answer: b) Uncover root motivations

23. "Synthesis" in the design process refers to

a) Collecting raw data b) Translating observations into actionable insights
c) Building the final product d) Selling the product

Answer: b) Translating observations into actionable insights

24. In an Empathy Map, the "Says" quadrant contains

a) The designer's opinions b) Direct quotes from the user
c) The user's hidden feelings d) The user's actions

Answer: b) Direct quotes from the user

25. A "Persona" is best defined as

a) A real person you interviewed
b) A fictional, generalized representation of an ideal target user
c) A celebrity endorser
d) The CEO of the company

Answer: b) A fictional, generalized representation of an ideal target user

26. The function of a Persona is to

a) Replace real users entirely
b) Make the abstract concept of "the user" concrete and relatable
c) Decorate the office wall
d) Impress the stakeholders

Answer: b) Make the abstract concept of "the user" concrete and relatable

27. "Ambitious Anna" in the text is an example of a

a) Real customer b) Persona
c) Brand mascot d) Employee

Answer: b) Persona

28. User "Pain Points" are

a) Physical injuries only
b) Specific frictions, frustrations, or failures a user experiences
c) The price of the product
d) The features the user likes

Answer: b) Specific frictions, frustrations, or failures a user experiences

29. Pain points can be functional, emotional, or

a) Imaginary b) Social c) Profitable d) Mathematical

Answer: b) Social

30. Which tool is often used to pinpoint where a user's experience dips below satisfaction?

a) A Balance Sheet b) Journey Mapping
c) A Gantt Chart d) A Pie Chart

Answer: b) Journey Mapping

31. "Problem Framing" focuses on

a) Jumping straight to the solution
b) Defining the core human problem before generating solutions
c) Finding the cheapest way to build something
d) Copying a competitor's idea

Answer: b) Defining the core human problem before generating solutions

32. "Problem Solving" (Traditional Approach) typically starts with

a) Deep empathy b) A pre-defined or assumed solution
c) User interviews d) Ethnographic research

Answer: b) A pre-defined or assumed solution

33. A well-framed Point of View (POV) statement typically follows which structure?

a) We need [Solution] because [Business Goal]
b) [User] needs [Need] because [Insight].
c) [Product] costs [Price]
d) [Company] wants [Profit].

Answer: b) [User] needs [Need] because [Insight].

34. In the example POV, "New online shoppers need a way to trust the final price immediately because..."

a) "...we want to make more money."
b) "...hidden fees make them feel manipulated."
c) "...shipping is expensive."
d) "...they are in a hurry."

Answer: b) "...hidden fees make them feel manipulated."

35. Framing the problem correctly ensures the Ideation stage focuses on

a) Maximizing user trust/value b) Minimizing milliseconds only
c) Using the newest technology d) Making the logo bigger

Answer: a) Maximizing user trust/value

36. Empathy prevents designers from building solutions based on

a) Real data b) Assumptions
c) User feedback d) Observations

Answer: b) Assumptions

37. Which of the following is an example of an "Emotional" pain point?

a) The app loads slowly b) The user feels stupid or anxious
c) The device is too heavy d) The battery dies quickly

Answer: b) The user feels stupid or anxious

38. Which of the following is an example of a "Social" pain point?

a) The product is too expensive b) The product creates an embarrassing situation
c) The instructions are unclear d) The buttons are too small

Answer: b) The product creates an embarrassing situation

39. Observational research focuses on what users _____, rather than what they say.

a) Think b) Do c) Buy d) Want

Answer: b) Do

40. Why is "Behaviour" often considered more truthful than verbal accounts?

a) Users never lie
b) Users may not be aware of their own adaptations or struggles
c) Behaviour is easier to record
d) Verbal accounts are always accurate

Answer: b) Users may not be aware of their own adaptations or struggles

41. The transition from Empathy to Define requires converting raw insights into

a) A focused challenge b) A finished product
c) A marketing plan d) A database

Answer: a) A focused challenge

42. Which stage of Design Thinking immediately follows Empathy?

a) Ideate b) Define c) Prototype d) Test

Answer: b) Define

43. Using simulated goggles to understand the visually impaired is a method of

a) Deep Interviewing b) Immersion
c) Surveying d) Focus Grouping

Answer: b) Immersion

44. A "Point of View" (POV) is essentially a

a) User Persona b) Problem Statement
c) Prototype d) Solution

Answer: b) Problem Statement

45. Empathy ensures that prototypes are kept

a) High-fidelity b) Low-fidelity and focused on experience
c) Expensive d) Secret

Answer: b) Low-fidelity and focused on experience

46. To build empathy, a designer must set aside their own

a) Tools b) Assumptions and biases
c) Creativity d) Budget

Answer: b) Assumptions and biases

47. "Radical transparency" in the text was identified as a need for users who

a) Abandon carts due to confusing shipping fees b) Want faster delivery
c) Want cheaper products d) Don't like online shopping

Answer: a) Abandon carts due to confusing shipping fees

48. In the text, a "touch-target optimized design" was the solution for

a) A farmer using a tractor
b) A healthcare worker wearing thick gloves
c) A visually impaired user
d) A busy parent

Answer: b) A healthcare worker wearing thick gloves

49. An Empathy Map helps the team understand the _____ the user makes.

a) Financial investments
b) Trade-offs
c) Products
d) Friends

Answer: b) Trade-offs

50. Ultimately, Empathy makes the design process_____

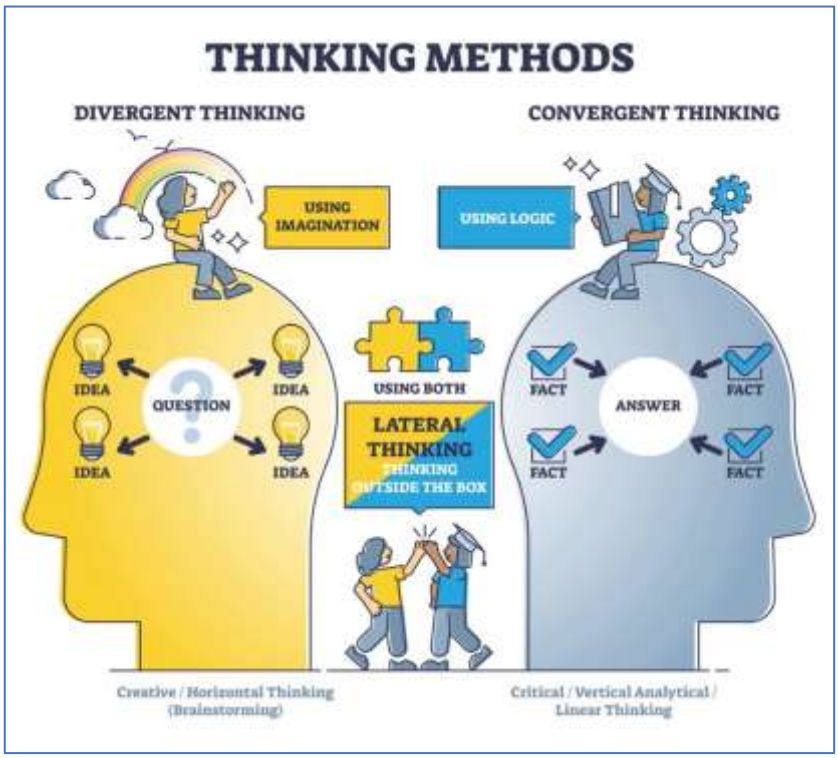
a) Faster
b) Human-centered
c) Cheaper
d) Automated

Answer: b) Human-centered

UNIT – 3

1. **The primary focus of the Ideation stage in Design Thinking is to**
   a) Define the problem statement b) Generate a large quantity of potential solutions
   c) Build the final product d) Test the product with users
   Answer: b) Generate a large quantity of potential solutions

2. **Which mode of thinking involves generating many different ideas from a single starting point?**
   a) Convergent Thinking b) Divergent Thinking
   c) Critical Thinking d) Linear Thinking
   Answer: b) Divergent Thinking



3. **What is the key mantra of Divergent Thinking?**
   a) Quality over quantity b) Feasibility first
   c) Quantity over quality d) Cost reduction
   Answer: c) Quantity over quality

4. **Convergent Thinking is primarily about**
   a) Opening up the solution space b) Deferring judgment
   c) Selecting and evaluating the best ideas d) Generating wild ideas
   Answer: c) Selecting and evaluating the best ideas

5. **What is the key mantra of Convergent Thinking?**
   a) Quantity and speed
   b) Quality and feasibility
   c) Novelty and chaos
   d) Diversity and volume
   Answer: b) Quality and feasibility

6. **Effective ideation requires alternating between**
   a) Divergence and Convergence
   b) Speaking and Listening
   c) Writing and Drawing
   d) Thinking and Sleeping
   Answer: a) Divergence and Convergence

7. **Which technique is described as a group method for generating a massive number of ideas quickly?**
   a) Dot Voting
   b) Mind Mapping
   c) Brainstorming
   d) Prototyping
   Answer: c) Brainstorming

8. **Which of the following is a strict rule of Brainstorming?**
   a) Criticize bad ideas immediately
   b) Focus on quality, not quantity
   c) Defer judgment
   d) Debate every suggestion
   Answer: c) Defer judgment

9. **In Brainstorming, "Wild Ideas" are**
   a) Discouraged
   b) Encouraged
   c) Ignored
   d) Forbidden
   Answer: b) Encouraged

10. **The phrase "Yes, and..." is used in Brainstorming to**
   a) Agree with the boss
   b) End the session
   c) Build on others' ideas
   d) Change the topic
   Answer: c) Build on others' ideas

11. Mind Mapping is best defined as a

a) Written list of pros and cons
b) Visual technique for structuring ideas around a central concept
c) Voting system
d) Spreadsheet analysis

Answer: b) Visual technique for structuring ideas around a central concept

12. In a Mind Map, where is the core problem or "How Might We" question placed?

a) In the bottom corner
b) In the centre
c) On the back of the page
d) It is not included

Answer: b) In the centre

13. SCAMPER is a checklist primarily used during which phase?

a) Prototyping
b) Testing
c) Divergent brainstorming
d) Final selection

Answer: c) Divergent brainstorming

14. In the acronym SCAMPER, what does 'S' stand for?

a) Simplify
b) Substitute
c) Select
d) Sell

Answer: b) Substitute

15. In SCAMPER, 'C' stands for:

a) Create
b) Copy
c) Combine
d) Cut

Answer: c) Combine

16. In SCAMPER, 'A' stands for

a) Adapt
b) Apply
c) Analyse
d) Avoid

Answer: a) Adapt

17. In SCAMPER, 'M' stands for

a) Make/Model b) Modify/Magnify c) Manage/Monitor d) Move/Mix

Answer: b) Modify/Magnify

18. In SCAMPER, 'P' stands for

a) Put to another use b) Print c) Prototype d) Plan

Answer: a) Put to another use

19. In SCAMPER, 'E' stands for

a) Evaluate b) Expand c) Eliminate/Minify d) Exit

Answer: c) Eliminate/Minify

20. In SCAMPER, 'R' stands for

a) Repeat b) Reverse/Rearrange c) Reject d) Reduce

Answer: b) Reverse/Rearrange

21. Using recycled plastic instead of wood for a chair is an example of which SCAMPER technique?

a) Combine b) Substitute c) Reverse d) Eliminate

Answer: b) Substitute

22. Integrating a light source into a chair is an example of

a) Substitute b) Eliminate c) Combine d) Adapt

Answer: c) Combine

23. Using a chair as an emergency flotation device is an example of

a) Modify b) Put to another use c)Reverse d) Adapt

Answer: b) Put to another use

24. Removing the backrest of a chair entirely is an example of

a) Eliminate b) Magnify c)Combine d) Rearrange

Answer: a) Eliminate

25. Changing the assembly process of a product so the user builds it from scratch is an example of

a) Substitute b)Reverse/Rearrange c)Minify d) Combine

Answer: b) Reverse/Rearrange

26. The "Four Categories" or "Post-It Grouping" method is used for

a) Generating ideas b) Initial convergence and grouping similar ideas

c) Interviewing users d) Designing the logo

Answer: b) Initial convergence and grouping similar ideas

27. The "Impact vs. Effort Matrix" plots ideas on which two axes?

a) Cost vs. Time b) Fun vs. Boring

c) Impact vs. Effort d) New vs. Old

Answer: c) Impact vs. Effort

28. In the Impact vs. Effort Matrix, ideas that are "High Impact, Low Effort" are called

a) Time Sinks b) Major Projects c) Fill-ins d) Quick Wins

Answer: d) Quick Wins

29. In the Impact vs. Effort Matrix, ideas that are "High Impact, High Effort" are called:

a) Quick Wins b) Major Projects c)Time Sinks d) Fill-ins

Answer: b) Major Projects

30. Which category of ideas in the Impact vs. Effort Matrix should generally be avoided?

a) Quick Wins b) Major Projects

c) Time Sinks (Low Impact, High Effort) d) Fill-ins

Answer: c) Time Sinks (Low Impact, High Effort)

31. "Dot Voting" is described as a

a) Solo activity, b) Rapid-fire democratic method for group selection

c) Computer algorithm d) Long debate process

Answer: b) Rapid-fire democratic method for group selection

32. What is the goal of Dot Voting?

a) To generate more ideas b) To narrow focus to ideas with highest collective support

c) To criticize ideas d) To assign blame

Answer: b) To narrow focus to ideas with highest collective support

33. Divergent thinking helps prevent the team from

a) Having too many ideas b) Settling on the first obvious solution

c) Spending too much money d) Working together

Answer: b) Settling on the first obvious solution

34. Convergent thinking uses criteria like

a) Desirability, Feasibility, Viability b) Colour, Shape, Size
c) Day, Month, Year d) Past, Present, Future

Answer: a) Desirability, Feasibility, Viability

35. A "Fill-in" idea in the Impact/Effort matrix is characterized by:

a) High Impact, High Effort b) Low Impact, Low Effort
c) High Impact, Low Effort d) Low Impact, High Effort

Answer: b) Low Impact, Low Effort

36. Mind Mapping helps teams explore:

a) Connections and tangents organically b) Financial spreadsheets
c) Employee performance d) Competitor pricing

Answer: a) Connections and tangents organically

37. Which SCAMPER technique involves adjusting a product to fit a different context?

a) Eliminate b) Adapt c) Reverse d) Substitute

Answer: b) Adapt

38. "Magnify" in SCAMPER can refer to:

a) Making the product smaller b) Removing features
c) Changing size, shape, or adding extra features d) Keeping it exactly the same

Answer: c) Changing size, shape, or adding extra features

39. Ideally, how many ideas should a brainstorming session aim for?

a) 5 b) 10 c) 100 (Quantity leads to variety) d) 1

Answer: c) 100 (Quantity leads to variety)

40. Deferring judgment means

a) Judging ideas later, not during generation b) Never judging ideas
c) Judging ideas immediately d) Asking a judge for legal advice

Answer: a) Judging ideas later, not during generation

41. Innovation requires ideas to be both

a) Cheap and Fast b) Novel and Relevant
c) Old and Reliable d) Complex and Difficult

Answer: b) Novel and Relevant

42. Which stage of Design Thinking precedes Ideation?

a) Prototype b) Test c) Define (and Empathy) d) Launch

Answer: c) Define (and Empathy)

43. The "How Might We" question is typically formulated during which stage?

a) Testing b) Define (transitioning to Ideation)
c) Prototyping d) Sales

Answer: b) Define (transitioning to Ideation)

44. If an idea is "High Effort" but "Low Impact," it is considered a:

a) Quick Win b) Time Sink c) Major Project d) Fill-in

Answer: b) Time Sink

45. Which tool visually groups related ideas on a wall?

a) Excel b) Post-It Grouping c)SCAMPER d) Dot Voting

Answer: b) Post-It Grouping

46. "Substitute" in SCAMPER could involve changing

a) The price b) A part, material, or process c) The logo d) The team members

Answer: b) A part, material, or process

47. The "R" in SCAMPER can stand for Reverse or:

a) Remove b) Rearrange c) Repeat d) Return

Answer: b) Rearrange

48. Why is "Quantity" encouraged in Brainstorming?

a) It looks busy b) Volume leads to variety and variety leads to quality options later

c) It uses more paper d) It confuses the team

Answer: b) Volume leads to variety and variety leads to quality options later

49. **Dot voting ensures that**:

a) Only the loudest person is heard b) Every team member has a voice

c) No decision is made d) The boss decides everything

Answer: b) Every team member has a voice

50. Ideation shifts the focus from "Defining" the problem to:

a) Solving the problem b) Ignoring the problem

c) Analysing the problem d) Complaining about the problem

Answer: a) Solving the problem

UNIT – 4

1. **The "Prototyping & Experimentation" stage involves transforming abstract ideas into:**
   a) Complex business plans
   b) Tangible artifacts
   c) Final products
   d) Marketing strategies
   Answer: b) Tangible artifacts

2. **Which core principle allows teams to quickly test assumptions?**
   a) Bias Towards Perfection
   b) Bias Towards Action
   c) Bias Towards Planning
   d) Bias Towards Analysis
   Answer: b) Bias Towards Action

3. **Prototyping is described not as building the final product, but as building a:**
   a) Vehicle for learning
   b) Source of revenue
   c) Marketing tool
   d) Legal requirement
   Answer: a) Vehicle for learning

4. **The primary goal of Low-Fidelity (Lo-Fi) prototyping is to:**
   a) Test aesthetics
   b) Rapidly test core assumptions and flow
   c) Creating a final polished look
   d) Write the final code
   Answer: b) Rapidly test core assumptions and flow

5. **Which of the following is a common material for Low-Fidelity prototypes?**
   a) Specialized software like Figma
   b) Metal and glass
   c) Paper and markers
   d) Final production plastics
   Answer: c) Paper and markers

6. **Low-Fidelity prototypes focus on**
   a) Aesthetics and polish
   b) Functionality and clarity
   c) Colores and fonts
   d) High-resolution images
   Answer: b) Functionality and clarity

7. **Why is the "rough" nature of a Lo-Fi prototype beneficial?**
   a) It saves money on software
   b) It encourages users to be more critical of the mechanics
   c) It looks artistic
   d) It takes longer to build
   Answer: b) It encourages users to be more critical of the mechanics

8. **High-Fidelity (Hi-Fi) prototypes are characterized by being:**
   a) Detailed representations closely resembling the final product
   b) Abstract sketches
   c) Made of cardboard
   d) Quick and disposable
   Answer: a) Detailed representations closely resembling the final product

9. Which tools are typically used for High-Fidelity prototyping?

a) Scissors and glue b) Figma or InVision c) Whiteboards d) Crayons

Answer: b) Figma or InVision

10. A primary focus of High-Fidelity prototyping is:

a) Basic information architecture b) Aesthetics and interactivity

c) Brainstorming, d) Rough concepts

Answer: b) Aesthetics and interactivity

11. One downside of High-Fidelity prototypes is that they are:

a) Too simple b) More time-consuming and expensive to adjust

c) Hard to visualize d) Disposable

Answer: b) More time-consuming and expensive to adjust

12. Storyboarding is best described as

a) A coding framework b) A visual storytelling tool like a comic strip

c) A 3D modeling technique d) A financial projection

Answer: b) A visual storytelling tool like a comic strip

13. Storyboards illustrate the user's experience in which sequence?

a) Start, Middle, End b) Input, Process, Output

c) Before, During, After d) Monday, Tuesday, Wednesday

Answer: c) Before, During, After

14. Storyboarding helps the team visualize

a) The code structure b) The cost of materials

c) When and why the product is used d) The exact dimensions of the product

Answer: c) When and why the product is used

15. "Sketching" is considered the _____ form of low-fidelity prototyping.

a) Most expensive b) Slowest c) Quickest d) Most complex

Answer: c) Quickest

16. Who can participate in sketching?

a) Only professional artists b) Only engineers

c) Everyone, regardless of artistic skill d) Only the CEO

Answer: c) Everyone, regardless of artistic skill

17. The emphasis of sketching is on

a) Precision and detail b) Speed and volume

c) Color accuracy d) Finalizing the design

Answer: b) Speed and volume

18. A "Mock-up" is typically:

a) Interactive and clickable b) A static, non-interactive visual representation

c) A physical cardboard model d) A video demonstration

Answer: b) A static, non-interactive visual representation

19. Mock-ups are primarily used to get feedback on:

a) Visual design and information hierarchy b) Physical weight
c) User flow logic d) Emotional context

Answer: a) Visual design and information hierarchy

20. A layered Photoshop file showing how a webpage looks is an example of a:

a) Sketch b) Wireframe c) Mock-up d) Storyboard

Answer: c) Mock-up

21. "Role-Playing" involves team members

a) Writing scripts b) Acting out the user-product interaction
c) Drawing cartoons d) Playing video games

Answer: b) Acting out the user-product interaction

22. Role-Playing is essential for prototyping:

a) Static websites b) Logos
c) Services or complex physical interactions d) Databases

Answer: c) Services or complex physical interactions

23. Which of the following can Role-Playing help discover?

a) Software bugs b) Logistical flaws and awkward physical interactions
c) Colour palette issues d) Font readability

Answer: b) Logistical flaws and awkward physical interactions

24. Rapid prototyping with simple materials emphasizes prioritizing

a) Fidelity over speed b) Speed of learning over fidelity
c) Aesthetics over function d) Cost over value

Answer: b) Speed of learning over fidelity

25. Paper and markers are ideal for testing

a) Weight and grip b) Screen flow and button placement
c) Ergonomics d) Emotional durability

Answer: b) Screen flow and button placement

26. Cardboard, Foam Core, and LEGOs are best used for:

a) Digital apps b) Physical models to test size and ergonomics
c) Writing code d) Graphic design

Answer: b) Physical models to test size and ergonomics

27. What makes Sticky Notes excellent for prototyping?

a) They are colourful b) They can be moved instantly to iterate on architecture
c) They are expensive d) They are permanent

Answer: b) They can be moved instantly to iterate on architecture

28. Why is "low investment" in materials important emotionally?

a) It makes the team attached to the design
b) It makes it easier to discard or change a failed prototype
c) It saves the company money
d) It looks professional

Answer: b) It makes it easier to discard or change a failed prototype

29. Prototyping allows teams to refine solutions before investing significant

a) Emotion b) Time or capital c) Paper d) Office space

Answer: b) Time or capital

30. The design process typically moves sequentially from

a) High-to Low-fidelity b) Low-to High-fidelity
c) Mid-to Low-fidelity d) High-to No-fidelity

Answer: b) Low-to High-fidelity

31. Hi-Fi prototypes allow teams to measure:

a) Task completion times and satisfaction b) The cost of paper
c) The durability of cardboard d) Team morale

Answer: a) Task completion times and satisfaction

32. Which technique is best for communicating ideas during the Ideation phase?

a) High-fidelity coding b) Sketching c) A/B Testing d) Market research

Answer: b) Sketching

33. An example of a Storyboard scenario given in the text is:

a) A user buying a car b) A user pre-ordering coffee during a commute
c) A user booking a flight d) A user fixing a leak

Answer: b) A user pre-ordering coffee during a commute

34. Mock-ups often show final:

a) Interactions b) Animations c) Colores, fonts, and layout d) Sounds

Answer: c) Colores, fonts, and layout

35. Role-playing a customer service call helps assess

a) The phone signal strength b) The script length
c) Emotional reactions and service flow d) The chair comfort

Answer: c) Emotional reactions and service flow

36. A "Wireframe" is mentioned as a type of

a) High-fidelity prototype b) Low-fidelity prototype
c) Final product d) Marketing brochure

Answer: b) Low-fidelity prototype

37. Hi-Fi prototypes provide a _____ feel.

a) Rough b) Sketchy c) Highly realistic d) Abstract

Answer: c) Highly realistic

38. Which prototyping method is best to test "Information Architecture"?

a) Role-Playing b) Sticky Notes / Lo-Fi c) Clay modelling d) Oil painting

Answer: b) Sticky Notes / Lo-Fi

39. Bias Towards Action ensures prototypes remain

a) Expensive b) Permanent c) Disposable d) Complex

Answer: c) Disposable

40. If a prototype fails during testing, the team should

a) Defend the design choice b) Hide the results

c) Discard or radically change it d) Launch it anyway

Answer: c) Discard or radically change it

41. Which feature is NOT a focus of Lo-Fi prototyping?

a) Polish b) Functionality c) Flow d) Clarity

Answer: a) Polish

42. Which tool is NOT listed as a Hi-Fi tool?

a) Figma b) InVision c) Cardboard d) Specialized software

Answer: c) Cardboard

43. Storyboarding compares the interaction to a

a) Novel b) Comic strip c) Movie script d) Painting

Answer: b) Comic strip

44. What does a "Before" panel in a storyboard typically show?

a) The solution b) The user's problem or frustration

c) The outcome d) The price

Answer: b) The user's problem or frustration

45. What does an "After" panel in a storyboard typically show?

a) The user feeling relaxed/efficient b) The user struggling

c) The login screen d) The settings menu

Answer: a) The user feeling relaxed/efficient

46. Sketching focuses on _____ and _____.

a) Quality and Detail b) Speed and Volume

c) Colour and Light d) Truth and Justice

Answer: b) Speed and Volume

47. Role-playing is useful for finding _____ physical interactions.

a) Smooth b) Awkward c) Perfect d) Invisible

Answer: b) Awkward

48. Using LEGOs to test user grip is an example of using

a) High-tech materials b) Simple materials c) Software d) Abstract theory

Answer: b) Simple materials

49. Prototyping gathers _____ user feedback.

a) Hypothetical b) Real c) Expensive d) Automated

Answer: b) Real

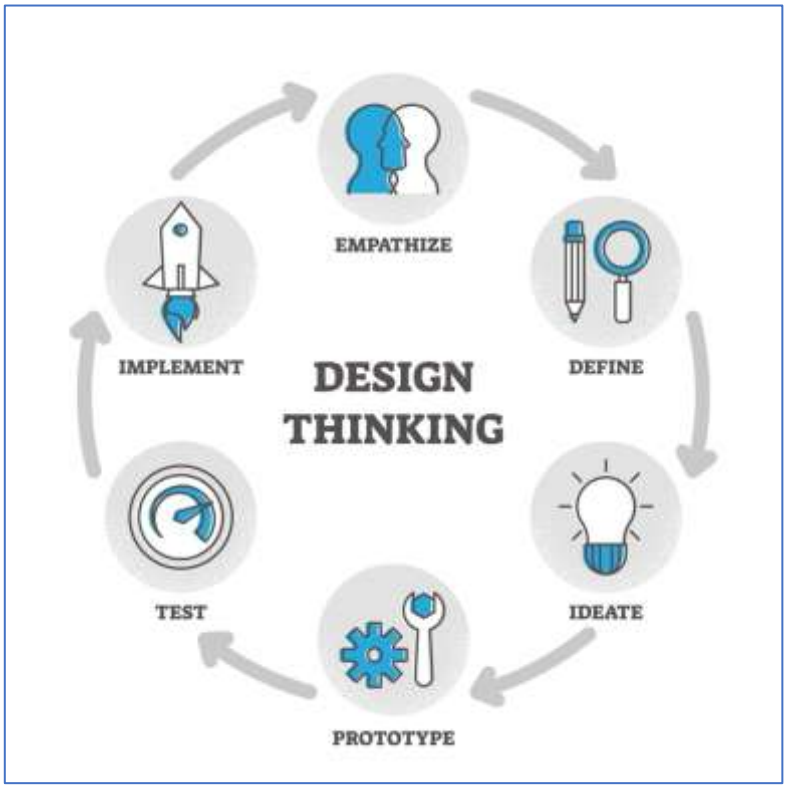
50. Ultimately, prototyping is a process of

a) Guessing b) Validating ideas c) Manufacturing d) Selling

Answer: b) Validating ideas

UNIT – 5

1. **The "Testing & Feedback" stage is primarily used to:**
   a) Generate new ideas b) Validate assumptions in the real world
   c) Secure funding d) Write code
   Answer: b) Validate assumptions in the real world



2. **Testing is deeply linked to which Design Thinking principle?**
   a) Iteration b) Ambiguity
   c) Divergence d) Linearity
   Answer: a) Iteration

3. **The goal of testing is to transform subjective ideas into:**
   a) Objective data on usability and desirability b) Financial spreadsheets
   c) Marketing slogans d) Employee performance reviews
   Answer: a) Objective data on usability and desirability

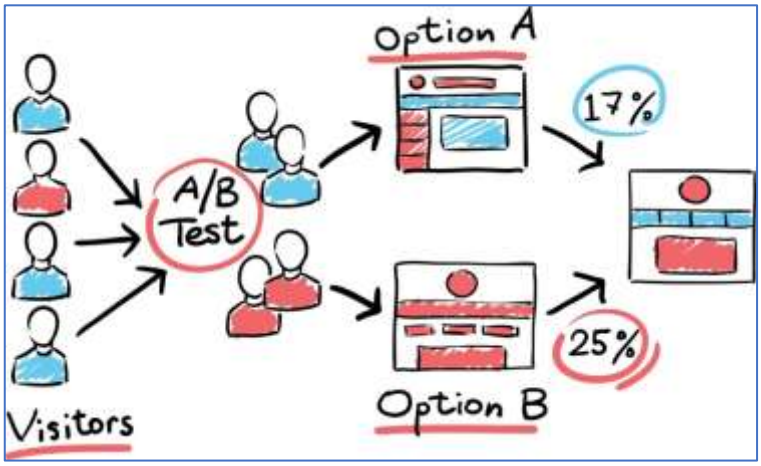
4. **What should be the designer's mindset during testing?**
   a) Defensive argumentation b) Empathetic inquiry
   c) Selling the product d) Perfecting the aesthetics
   Answer: b) Empathetic inquiry

5. **The fundamental objective of testing is NOT to prove perfection, but to**
   a) Identify failures and flaws early b) Convince the user to buy
   c) Show off the design skills d) Make the prototype look expensive
   Answer: a) Identify failures and flaws early

6. **Which testing method involves giving users specific tasks like "Find the shipping cost"?**
   a) Guerilla Testing b) A/B Testing
   c) Usability Testing (Task-Based) d) Focus Groups
   Answer: c) Usability Testing (Task-Based)

7. **"Guerilla Testing" is best described as:**
   a) Testing in a lab with scientists b) Quick, informal testing in public places
   c) Testing with military personnel d) Testing only with competitors
   Answer: b) Quick, informal testing in public places

8. **A/B Testing is primarily used for:**
   a) Generating wild ideas b) Testing subtle iterative changes/optimization
   c) Creating a new business model d) Interviewing users for the first time
   Answer: b) Testing subtle iterative changes/optimization



9. **In the text, a user clicking a static image instead of a button revealed a flaw in:**
   a) The internet connection b) Visual cues
   c) The price d) The product name
   Answer: b) Visual cues

10. **Every struggle a user faces during testing invalidates:**
   a) The user's intelligence b) An assumption the designer made
   c) The business plan d) The testing software
   Answer: b) An assumption the designer made

11. **Insights from testing are often synthesized into which section?**
   a) "What we learned" b) "Why we are right"
   c) "Budget analysis" d) "Competitor list"
   Answer: a) "What we learned"

12. **In the hospital kiosk example, the design flaw was a lack of empathy for:**
   a) Children b) Physical constraints/mobility issues
   c) Doctors d) Technology
   Answer: b) Physical constraints/mobility issues

13. In the budgeting app example, users entered the wrong date because of

a) An efficiency flaw (field too small) b) Lack of knowledge
c) A system crash d) Slow internet

Answer: a) An efficiency flaw (field too small)

14. In the water bottle example, the team added ergonomic ridges because

a) It looked cooler b) Users complained their fingers slipped (grip issue)
c) It was cheaper to manufacture d) The bottle was leaking

Answer: b) Users complained their fingers slipped (grip issue)

15. Scaling is the process of

a) Expanding a successfully piloted concept b) Measuring the weight of a product
c) Reducing the team size d) Ignoring feedback

Answer: a) Expanding a successfully piloted concept

16. An idea is not truly an innovation until it is

a) Written down b) Successfully implemented and adopted at scale
c) Patented d) Sketched

Answer: b) Successfully implemented and adopted at scale

17. Before scaling, a prototype must prove Desirability, Feasibility, and

a) Popularity b) Viability
c) Complexity d) Beauty

Answer: b) Viability

18. "Process Standardization" during scaling involves

a) Making the process dependent on the original designer
b) Stripping dependence on the original team (documentation/training)
c) Hiring more designers
d) Increasing the budget randomly

Answer: b) Stripping dependence on the original team (documentation/training)

19. What is identified as the single greatest barrier to scaling?

a) Technology b) Resistance to change (Cultural buy-in)
c) Lack of office space d) Too many ideas

Answer: b) Resistance to change (Cultural buy-in)

20. "Financial Metrics" measure

a) Customer happiness b) Internal speed
c) Economic return and sustainability (Viability) d) Employee satisfaction

Answer: c) Economic return and sustainability (Viability)

21. "Return on Innovation Investment" (ROI) compares profit generated to:

a) The number of employees b) The total investment (R&D, prototyping, scaling)
c) The number of prototypes d) The competitor's profit

Answer: b) The total investment (R&D, prototyping, scaling)

22. "Cost Reduction" is often achieved through:

a) Product Innovation b) Process Innovation
c) Social Innovation d) Marketing Innovation

Answer: b) Process Innovation

23. "Customer Metrics" primarily gauge

a) Feasibility b) Desirability (Market acceptance)
c) Viability d) Profitability

Answer: b) Desirability (Market acceptance)

24. "Adoption Rate" measures

a) How much money a customer spends
b) The speed and breadth of user incorporation
c) How many employees use the product
d) The speed of manufacturing

Answer: b) The speed and breadth of user incorporation

25. "Net Promoter Score" (NPS) measures

a) Customer loyalty and willingness to recommend
b) The net profit of the company
c) The number of returns
d) The speed of the website

Answer: a) Customer loyalty and willingness to recommend

26. "Operational Metrics" assess

a) The customer's feelings
b) The internal efficiency and speed of the innovation process
c) The stock market price
d) The social impact

Answer: b) The internal efficiency and speed of the innovation process

27. "Time-to-Market" is defined as

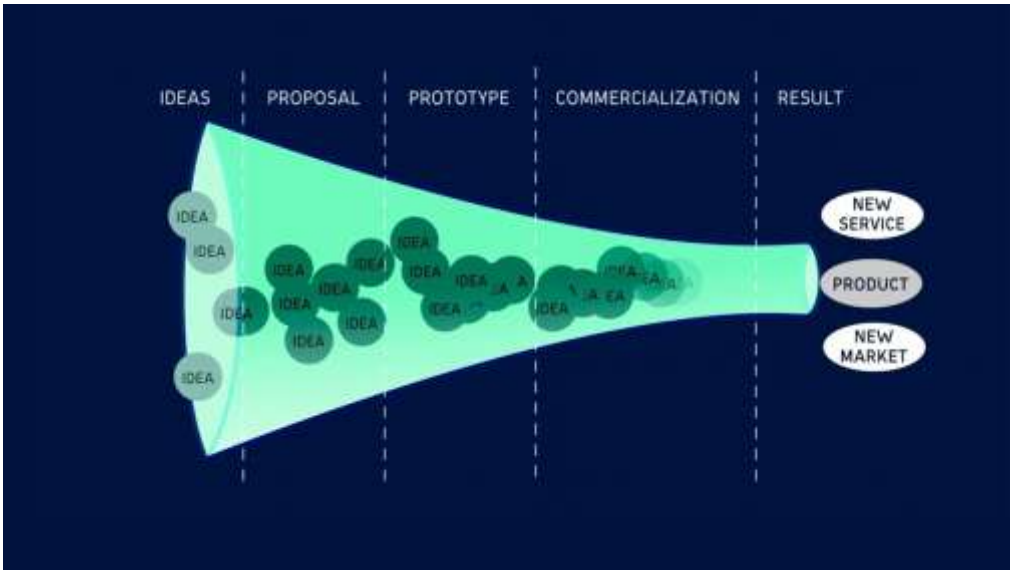
a) The time shop opens
b) The speed from idea generation to commercial launch
c) The duration of a user interview
d) The length of the marketing campaign

Answer: b) The speed from idea generation to commercial launch

28. "Innovation Pipeline Health" tracks

a) The physical health of employees
b) The number of ideas moving through the innovation funnel
c) The plumbing of the office building
d) The number of rejected ideas

Answer: b) The number of ideas moving through the innovation funnel



29. Which organizational culture trait is the most significant barrier to innovation?

a) Open communication b) Fear of failure
c) Diversity d) Flexible hours

Answer: b) Fear of failure

30. "Resource Constraints" refer to insufficient allocation of

a) Desks and chairs b) Time, funding, or personnel
c) Coffee and snacks d) Internet bandwidth

Answer: b) Time, funding, or personnel

31. Innovation efforts must be clearly linked to what to avoid being defunded?

a) The CEO's hobby b) The company's overall business strategy
c) The latest social media trend d) The competitor's strategy

Answer: b) The company's overall business strategy

32. "Silos" in an organization prevent

a) Noise pollution b) Cross-pollination of knowledge and collaboration
c) Data theft d) Overworking

Answer: b) Cross-pollination of knowledge and collaboration

33. Social and environmental challenges are often called

a) Easy Problems b) Wicked Problems
c) Simple Problems d) Math Problems

Answer: b) Wicked Problems

34. "Wicked Problems" are characterized by being

a) Easy to solve b) Difficult to define and interconnected
c) Unimportant d) Purely financial

Answer: b) Difficult to define and interconnected

35. Design Thinking shifts sustainability from a regulatory issue to a

a) Desirability issue b) Legal issue
c) Technical issue d) Political issue

Answer: a) Desirability issue

36. A "Systemic View" encourages designing

a) Single disposable products b) Entire service ecosystems (e.g., circular economy)
c) Logos only d) Isolated features

Answer: b) Entire service ecosystems (e.g., circular economy)

37. In the context of sustainability, empathy is extended to

a) Shareholders only
b) Non-traditional stakeholders like the environment and future generations
c) Only current paying customers
d) Competitors

Answer: b) Non-traditional stakeholders like the environment and future generations

38. The loop between Testing and Prototyping/Ideation is

a) A waste of time b)Critical for closing the learning loop
c) Optional d) Only for software

Answer: b) Critical for closing the learning loop

39. If a user says "I can't find the menu," this reveals a need for

a) Better usability/clarity b) More features
c) Less testing d) A new colour scheme

Answer: a) Better usability/clarity

40. If a user asks "Where is the search bar?", this reveals a need for

a) Simplification b) New features (What was missing?)
c) Removal of features d) A/B testing

Answer: b) New features (What was missing?)

41. Adaptation of "Infrastructure" during scaling might involve

a) Buying new furniture b) Updating the technology stack or supply chain
c) Painting the walls d) Changing the company name

Answer: b) Updating the technology stack or supply chain

42. "Customer Lifetime Value" (CLV) is a

a) Financial Metric b) Customer Metric
c) Operational Metric d) Social Metric

Answer: b) Customer Metric

43. A risk-averse environment stifles:

a) Payroll b) Iteration and Experimentation
c) Documentation d) Compliance

Answer: b) Iteration and Experimentation

44. "Gamifying" water usage to make it appealing is an example of

a) Human-Centered Sustainability b) Corporate Greed
c) Regulatory Compliance d) Guerilla Testing

Answer: a) Human-Centered Sustainability

45. Testing helps identifying what was "unnecessary" so it can be

a) Expanded b) Simplified or removed
c) Hidden d) Sold separately

Answer: b) Simplified or removed

46. A/B Testing involves comparing

a) Two completely different products b) Two variations of a high-fidelity prototype
c) A sketch and a final product d) An idea and a business plan

Answer: b) Two variations of a high-fidelity prototype

47. The result of the "Testing" phase is

a) A final guarantee of success b) Learning that fuels the next iteration
c) A patent d) A sales contract

Answer: b) Learning that fuels the next iteration

48. Which metric assesses the "Feasibility" aspect of innovation?

a) NPS b) Time-to-Market
c) ROI d) Adoption Rate

Answer: b) Time-to-Market

49. Successful scaling requires the solution to be stripped of

a) All features b) Its dependence on the original design team
c) Its user interface d) Its value

Answer: b) Its dependence on the original design team

50. "What we learned" directly informs

a) The previous step b) The next steps (Iteration)
c) The HR department d) The legal team

Answer: b) The next steps (Iteration)